Activity Browser Setup Guide

Install Activity Browser • Import ecoinvent • Install Brightway (for sensitivity analysis)

This guide standardizes the installation workflow for Activity Browser (AB), importing the ecoinvent database, and installing Brightway in a separate environment for sensitivity analysis.

Before you start

Prerequisites (before installing Activity Browser)

- A Windows PC with internet access and permission to install software (or IT self-service access).
- An ecoinvent account with access to the database version you want to use.
- At least ~5–10 GB free disk space (more is better, depending on ecoinvent version).

1) System requirements

- **Operating system:** Windows 10/11 (recommended), or macOS/Linux
- **Admin rights:** Permission to install software (or access to your institute IT self-service installer)
- **Disk space:** Minimum **5–10 GB free** (ecoinvent import can require more)
- **RAM: 8 GB minimum**, **16 GB recommended** (especially for importing ecoinvent)
- **Internet:** Stable connection for downloading packages and databases

2) Software requirements

- Anaconda or Miniconda installed (needed for `conda`)
- Anaconda Prompt / Terminal access (to run conda commands)
- (Recommended) Git or GitHub Desktop if you want to manage versions and push to GitHub

3) Data / account requirements

- ecoinvent account with access to the version you plan to import (e.g., ecoinvent 3.10/3.11/3.12)
- Know what you need:
  - Database version (e.g., 3.11)
  - System model (e.g., consequential / cut-off / allocation at point of substitution)

Step 1 — Install Anaconda (or Miniconda)

1. On INSA-managed PCs: open **Self Service Logiciels (WAPT)** and search **Anaconda Python 3**.
2. Install Anaconda (installation can take ~20 minutes).
3. If your laptop has limited space or Anaconda is not supported, install **Miniconda** instead (lighter, command-line only).

Why: Activity Browser is distributed via **conda** (package manager), so you need Anaconda or Miniconda.

Step 2 — Install Activity Browser (AB)

Follow the official installation guide: https://github.com/LCA-ActivityBrowser/activity-browser/wiki/Installation-Guide

4. Open **Anaconda Prompt** (Start menu → search "Anaconda Prompt").
5. Add the **conda-forge** channel (recommended for AB and dependencies).
6. Create a dedicated conda environment called **ab** and install Activity Browser inside it.
7. Activate the environment and start AB.

```
conda config --prepend channels conda-forge


conda create -n ab -c conda-forge activity-browser
# When asked: Proceed [y]/n → type y and press Enter


conda activate ab
activity-browser
```

After the first launch, AB will open and you can configure projects and databases.

Update Activity Browser (recommended periodically)

```
conda activate ab
conda update activity-browser
```

Step 3 — Import ecoinvent into Activity Browser

There are two common ways to import ecoinvent into AB. Use **Option A** if you want AB to manage the download/import. Use **Option B** if you prefer downloading ecoSpold files yourself and importing them manually.

Option A — Import directly inside AB (recommended)

8. Open **Activity Browser**.
9. Go to **Databases** → **Set up your project with default data**.

10. Select **ecoinvent** and **biosphere3**.
11. Click **Login** and enter your ecoinvent credentials: <your_username> / <your_password>.
12. Select the required **version** and **system model** (e.g., *ecoinvent 3.11 consequential*).
13. Start the import and wait until it completes (time depends on PC and internet connection).

Option B — Download ecoSpold and import manually

14. Go to the official ecoinvent website and log in: https://ecoinvent.org/
15. Navigate to **Resources** and locate the dataset you need.
16. Download the relevant **ecoSpold02** archive (often a .7z file). Example: *ecoinvent 3.11 consequential ecoSpold02*.
17. In Activity Browser, use the database import tool to import the downloaded ecoSpold files.

Tip: Keep the downloaded archive in a clearly named folder (e.g., D:\LCA\ecoinvent\3.11_consequential\) so you can reuse it.

Step 4 — Install Brightway in a separate environment (for sensitivity analysis)

To avoid dependency conflicts, install Brightway in its own conda environment. Below is a stable setup using Python 3.9.

18. Open **Anaconda Prompt**.
19. Create a new environment (example name: brightway2).
20. Activate it.
21. Install Brightway via pip.
22. Verify the installation.

```
conda create --name brightway2 python=3.9
# Proceed [y]/n → type y and press Enter

conda activate brightway2
pip install brightway2

python -c "import brightway2 as bw; print(bw.__version__)"
```

Troubleshooting (quick fixes)

- **Conda command not found:** Reinstall Anaconda/Miniconda or ensure it is added to PATH.
- **Environment activation fails:** check environments with `conda env list`.

- **AB doesn’t start:** run `conda activate ab` then `activity-browser` and read the error message; often a missing dependency can be fixed with `conda update --all`.
- **ecoinvent import fails:** confirm you selected the correct version/system model and that you have access rights in your ecoinvent account.